

one!), and bureaucratic meetings.

Since the climate insouciance of my generation will be visited upon our children's, and theirs, I feel it is imperative to bring in awareness in schools. The government's Atal Innovation Mission's Atal Tinkering Labs—10,000 in high schools across the length and breadth of India—provide a hands-on, popular, and fun science-based setting within which kids' heat awareness can be enhanced.

And if one wants awareness to spread among the less-included segments of society, is it feasible to solicit the assistance of our amazing, albeit overworked, ASHA health workers, and other grass-roots organisations, to spread literacy about taking the damaging effect of heat on the body seriously?

Information: We are, of course, inundated by data. But data isn't the same as information. We need the latter to be useful and actionable. For example, early warning systems and adaptation plans are much more

useful if these can capture, not just aggregates, but also how ambient heat is affecting individuals. As my colleague, Dr Satchit Balsari at Harvard Medical School put it to us, "Saying 1.5° C excess temperature today is like asking a physician on morning rounds: How is the entire hospital feeling today? It's an unhelpful number that does not communicate the reality of billions of people most at risk."

A large-scale research project that I co-lead at Harvard, with climate scientists and medical doctors, done collaboratively with the redoubtable women's trade union, SEWA, made the absence of relevant information blindingly clear, so I'll use it as an illustration. In our project, we wanted to understand the effect of heat stress on some of the most vulnerable members of society—informal workers represent as much as 90% of the workforce of most developing countries. SEWA members agreed to a long-term plan to work with us. SEWA women are involved in agriculture, home-based informal work,

salt pan tending, bangle making, running *kirana* stores, you name it, providing variation in setting.

Even with seemingly-ubiquitous data sources, the requisite information is entirely missing. Consider now commonplace satellite data. But satellite imagery has insufficient resolution for our purposes—generally available at 100–1,000 m, but not at individual human scales. In our study, we want conditions under a roof or just above ground, but satellites sense upper-most solid surfaces. And satellite data is usually available once daily, we need 24/7 data streams to understand how heat affects the body.

So, over the past several months, our team has built an infrastructure to measure individual exposure to heat, long-term (over 365 days), 24/7 measurements (that is, in the home and outside it) along with a range of measures of body function.

Our large-scale data collection has now ensued. Pilot studies already show, for example, that despite night-time summer temperatures cooling to say 27° C, indoor temperatures remain high due to various micro-climatic determinants (ventilation, construction materials, thermal lag), which means that SEWA women are out in the heat during the day and come back to heat all night. One hopes that, with better information comes better ability to design systems to mitigate effects of always-on heat. This sort of understanding is especially relevant in rapidly growing economies like India; the Ministry of Power estimates that 40% of the building stock that will exist two decades from now is yet to be built.

As another illustration of available data that needs some work to translate into usable information,



GIVEN THE UNPREDICTABLE NATURE OF CLIMATE DISASTERS, SOCIETY HAS TO CREATE A CAPABILITY TO RESPOND QUICKLY